

Design and Development of a Web-Based Tracer Study System in Higher Education Using the Waterfall Method (Case Study at Dian Nusantara University)

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ABSTRACT

This study aims to design and develop a web-based tracer study system at Dian Nusantara University using the Waterfall method. The system is designed to assist higher education institutions in tracking alumni's career paths and their level of satisfaction with the education they received. Additionally, this system aims to expedite the survey form completion process by alumni, ensuring that the collected data is more accurate and easier to analyze. The development stages include requirements analysis, design, implementation, and testing to ensure system performance. The outcome of this research is a platform that allows alumni to fill out surveys online, facilitating data processing by the university. This system is expected to strengthen the relationship with alumni and provide valuable information for curriculum development and educational planning.



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INTRODUCTION

The success indicator of higher education is how relevant graduates are to the needs of the job market. Higher education institutions are expected to produce graduates who are competitive and ready to contribute to development, both locally and globally. Although many universities have conducted tracer studies on their graduates, the results have not been satisfactory. Data from tracer studies are crucial for universities to evaluate the effectiveness of their education on students. In fact, during grant processes and accreditation, data from tracer studies, such as the waiting period for graduates, the percentage of graduates who are employed, and their initial income, are often required. [4]

Currently, Dian Nusantara University (UNDIRA) does not have a dedicated tracer study application. The alumni data tracking mechanism is done by contacting alumni via email and attaching a Google Form as a medium for filling out the survey. Although Google Forms is easy to use and free, there are several limitations that Dian Nusantara University must consider for the sustainability of its tracer study activities in the coming years.[2]

The proposed solution is the development of a Tracer Study application. The purpose of developing the tracer study application is to integrate the research needs of the tracer study into an application. This application offers significant benefits, such as helping study programs improve their curriculum by providing feedback and preparing the necessary tracer study documentation for accreditation processes. Through this platform, alumni career tracking can be done more efficiently and systematically. Alumni can be invited to fill out the survey online, reducing barriers such as difficulty in contacting them and receiving responses. Additionally, the web-based tracer study allows for centralized data storage and management, making it easier for relevant parties, such as study program staff and education decision-makers, to access and analyze the data. Therefore, creating a web-based tracer study can be an effective step in improving the accuracy and relevance of information about alumni career paths and supporting efforts to evaluate and develop educational programs at higher education institutions . [7]

The main issue faced is the collection of alumni data at Dian Nusantara University, especially related to the evaluation of graduate success in the information technology industry. Manual data collection results in difficulties in tracking alumni career paths, low response rates, and challenges in storing and analyzing data adequately. The proposed solution is the creation of a web-based tracer study platform that allows for online data collection. With this platform, it is expected that the process of tracking and collecting alumni data will be more efficient and structured. Furthermore, centralized data storage and easy access for relevant parties will facilitate data analysis and decision-making related to the enhancement of educational programs and curricula .[3]

METHODS

Data Collection

In this study, data collection techniques were conducted using interviews and observations. The purpose of the interviews was to gather relevant data and information to assist in observing and analyzing the processes occurring in the tracer study system at Dian Nusantara University. The data and information used were sourced from books, scholarly articles, and journals available on the internet. The observation aimed to collect existing data and information.[1]

Software Development Methods

This information system was developed using the SDLC (Software Development Life Cycle) model, which includes problem analysis, requirements specification, design, coding, testing, system integration, and operation. System maintenance and evolution are necessary to ensure the continued functionality of the system. The advantage of applying the fountain model is that certain stages can be prioritized or skipped, although some stages, such as implementation, cannot be prioritized before the design phase. The data collection methods used were observation and interviews. The system testing results indicate that the mobile-based tracer study information system enables higher education institutions to effectively and efficiently collect the required information.[5]

Waterfall Method

To carry out the software development process, the method used is the waterfall method. The Waterfall method is a sequential method of each development part that is owned. The Waterfall method is a development method that is a Linear Sequential Model which is often also known as the classic life cycle, starting from the initial part of the system development specification, namely the planning, modeling, construction and delivery of the system. to users (deployment), which ends with support for the complete software produced. [6]The development method uses a waterfall model similar to that in the following picture:

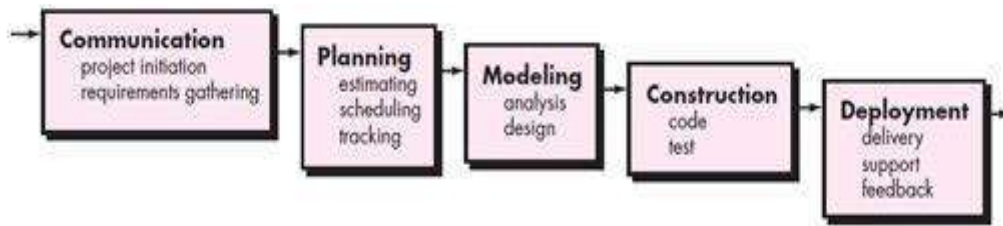


Figure 1. Development of the Waterfall method

This waterfall method has several steps:

1. **Communications**
This stage is used to analyze the importance of the tracer study system, as well as the steps in collecting data and holding meetings with research objects, both to collect additional data and those contained in journals or articles.
2. **Planning**
In this planning stage, which is a continuation of the communication stage, the system's services, constraints, and objectives are developed through consultation with system users. The description of the services provided by the system and the restrictions specified in detail serve as a detailed component of the system.
3. **Modeling**
This stage is a design and modeling stage that aims to translate system requirements, focusing on designing data structures, software architecture, interface appearance, and program algorithms. The goal is to gain a better understanding of the overall picture of what will be done.
4. **Construction**
In the Construction process, this is the process of translating the design form into a code or form of language that can be read by a computer. The programmer will translate the code into a software interface through coding. This process is the real process of handling software, therefore the use of computers is maximized in this process. After the coding is complete, testing is carried out on the system and also the code that has been created. The goal is to find errors that may occur so that they can be corrected later.
5. **Deployments**
The Deployment process is the stage of implementing the software to users, or it could also be said to be the final stage in creating the software. After analysis, design and coding have been carried out, the system that has been created will be used by users. Then the software that has been created will always be checked periodically, software repaired, software evaluated, and software developed based on the feedback provided so that the system can continue to run and develop according to its function.

Unified Modeling Language (UML)

Unified Modeling Language, hereinafter referred to as (UML), is a system development technique that uses graphical language as a tool for documenting and carrying out system specifications. In this research, the system design uses UML, which consists of creating use case diagrams, activity diagrams, and sequence diagrams. System software design includes database design, producing a mapping of tables depicted with an Entity Relationship Diagram. [13]

Test Result

System testing uses Blackbox testing, which is a software testing technique that focuses on the functional specifications of the software without having to pay attention to software details. Black box testing works by entering data in each form. Black box testing allows software

developers to create a set of input conditions that will train all the functional requirements of a program so they can find out whether the program can run as desired or not. The advantage over the black box testing method is that the tester does not have to have more knowledge of a particular programming language because this testing is carried out from the user's perspective to help reveal inconsistencies in the requirements specification. [7]

RESULTS AND DISCUSSION

Analysis of the tracer study system using Google Forms for collecting alumni data reveals several weaknesses, including low response rates and difficulties in data management. The system lacks a reminder feature for alumni who have not completed the form and does not provide centralized data storage. The development of a web-based application for the tracer study enables alumni to access a portal and complete the survey online more flexibly, which is expected to improve response rates. Additionally, all collected data will be stored in a centralized database, facilitating easier access and analysis. This process can also be monitored directly by administrators, both in terms of the number of alumni responding and the results of each survey question.

Analysis

The development of a website for alumni form completion serves as an effective solution to address the challenges in collecting tracer study data. This website is designed with several key features, including survey question formulation that adheres to the regulations of the Ministry of Education and Culture (Kemendikbud) and additional inputs from the university to enhance the system. The website also includes the creation of an alumni database for broadcasting needs and data management. The survey mechanism simplifies the process by tailoring it according to alumni employment status. Additionally, the website is equipped with a user-friendly tracer study form page, a dashboard for data analysis, and an export function that complies with Kemendikbud's formatting requirements.

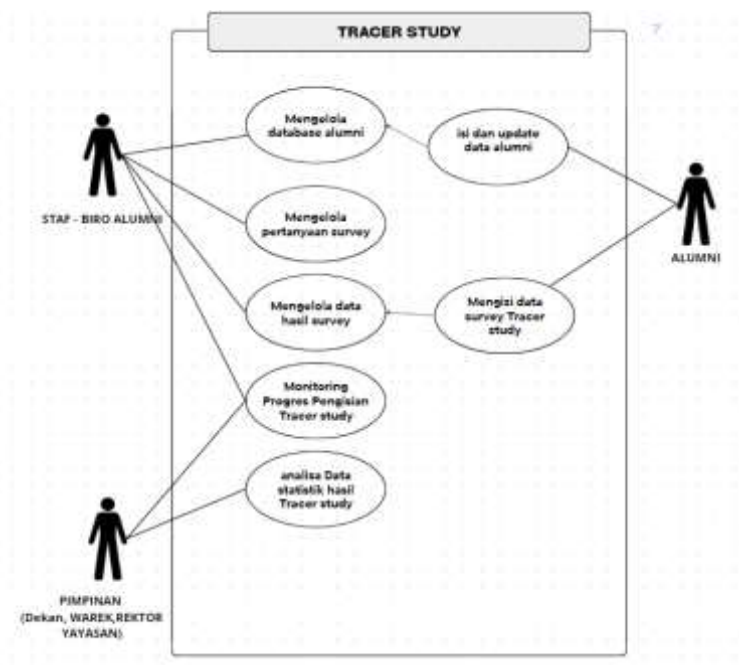


Figure 2. Use Case Diagram

Use Case Diagrams

The use case designed for the alumni and staff modules in the tracer study includes staff roles aimed at performing five tasks: managing alumni data, study programs, survey questions,

survey results, and survey reports. Meanwhile, the alumni role is designed to update personal information and complete the tracer study survey.

Implementation

a. Main page

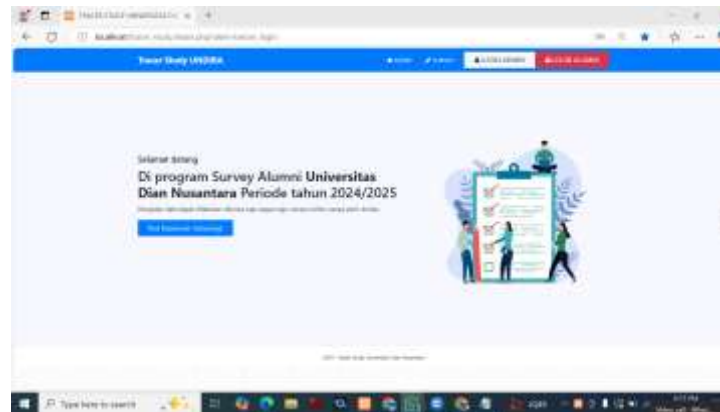


Figure 3. Front page (Part 1)

b. Admin Page

Figure 3 shows the front page which is the first page that visitors see when entering the website.



Figure 4. Front page (Part 2)

Figure 4 The following is the admin dashboard page, which displays information on the number of alumni, survey completion status, a graph of tracer study survey results, the ability to manage survey questions, and the option to send messages to alumni via email blast.

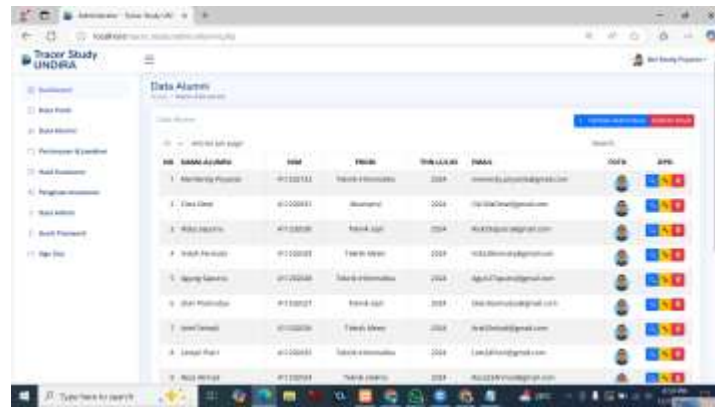


Figure 5. Alumni Data Page

Figure 5 This page features functions for adding alumni data and editing alumni personal information. Additionally, it includes a blast email function to send email messages to alumni, reminding them to complete the tracer study survey.

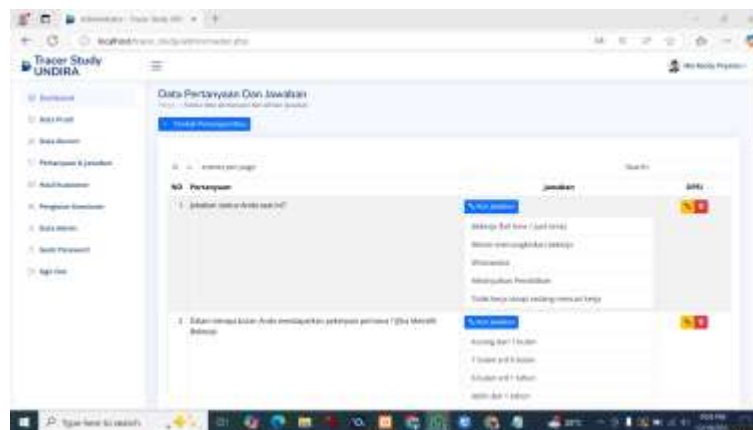


Figure 6. Tracer Study question edit page

Figure 6 On this page, the admin can modify the questions and answers, which can then be sent to the alumni.

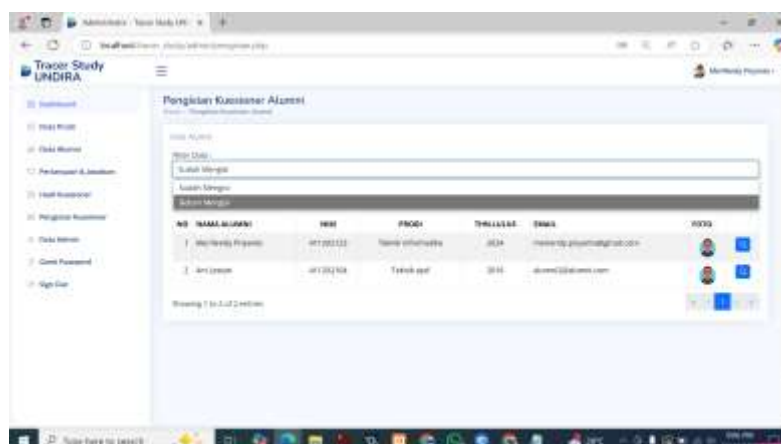


Figure 7. The Tracer Study Completion Monitoring Page

Figure 7 On this page, the admin can view in detail which alumni have completed the survey and which have not.

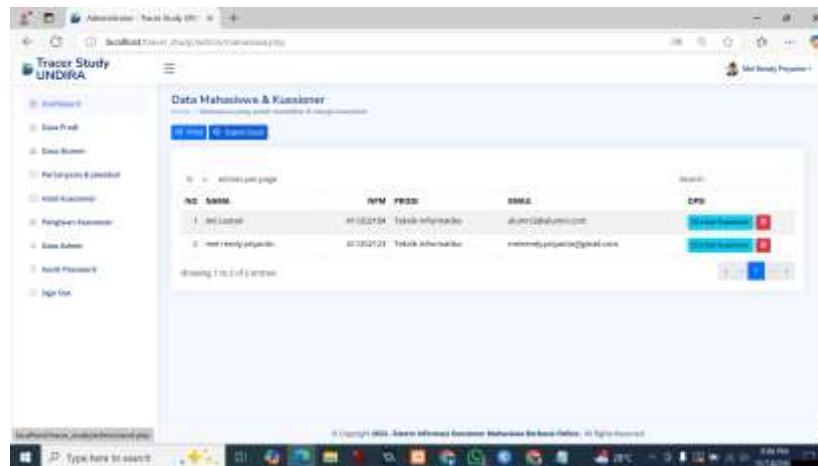


Figure 8. Tracer Study Survey Results Page

Figure 8 On this page, the admin can view the survey results and export them into an Excel format.

c. Login Alumni



Figure 9. Login Alumni Page

Figure 9 On this page, alumni log in using a combination of their username and NIM (Student Identification Number) password.

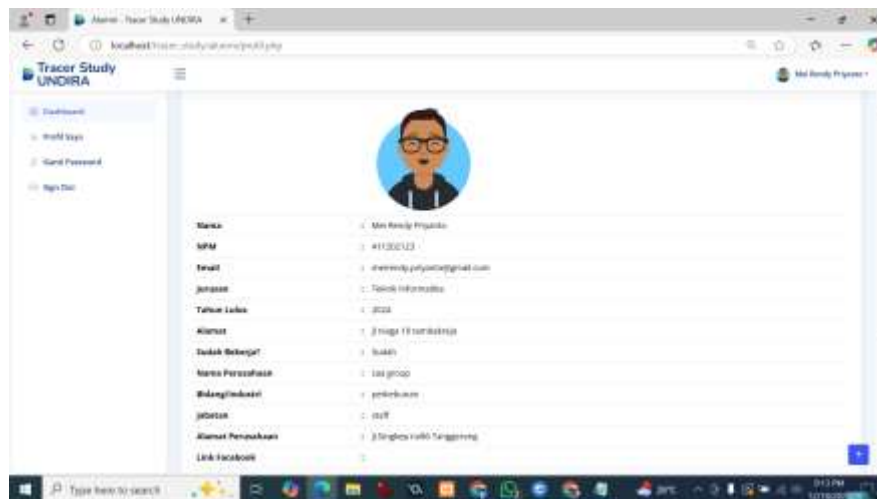


Figure 10. The Alumni Personal Data Page

Figure 10 On this page, alumni can update their personal information, such as the company they work for, the city they live in, contact details, and social media profiles.

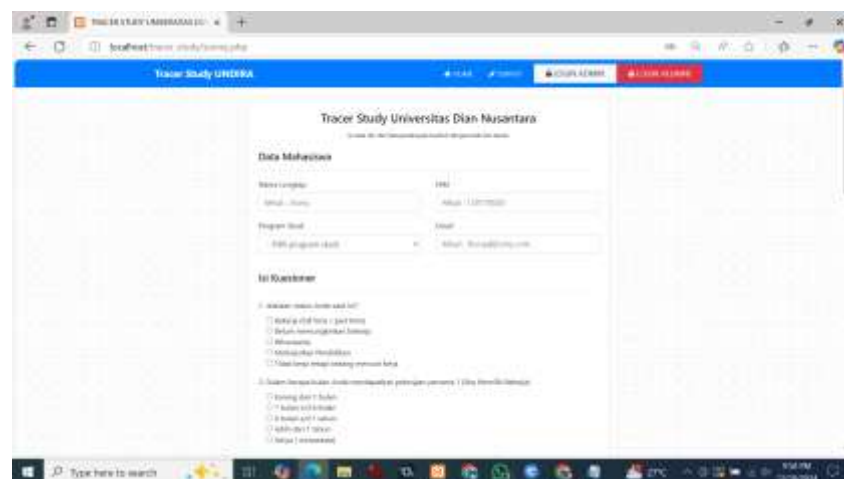


Figure 11. Alumni survey completion page

System Testing

With the requirements specifications already outlined before the system is ready for use, system testing is carried out first. The following are the results of the testing system that has been carried out on website functions

Table 1. System Test Results Table

No	Pengujian	test case	hasil yang diharapkan	hasil pengujian	kesimpulan
1.	Login ke halaman admin	Pengisian username dan password	masuk ke halaman admin/dashboard	95%	valid

2.	Logout dari halaman admin	klik tombol logout pada navbar sebelah kanan atas	keluar dari halaman admin dan masuk ke halaman login pengisian username dan password	95%	valid
3.	masuk ke halaman alumni	CRUD pada data alumni	respon terhadap data baik penambahan dan perubahan data alumni dapat dijalankan dengan baik	95%	valid
4.	blast email	menirim email ke alumni	alumni dapat menerima pesan pengisian tracer study lewat email	95%	valid
5.	masuk ke halaman Pertanyaan dan jawaban survey	CRUD pada pertanyaan dan jawaban	data pertanyaan dan jawaban responsif sesuai fungsi yang diharapkan	95%	valid
6.	masuk ke halaman monitoring pengisian tracer study	memilih status pengisian (sudah/belum) pada daftar alumni	dapat menampilkan data alumni yang sudah mengisi dan belum mengisi tracer study	95%	valid
7.	masuk ke halaman hasil tracer study	melihat data hasil pengisian dan export ke dalam format Microsoft excell	dapat export data ke format ms.excell	95%	valid
8.	masuk ke halaman admin	CRUD pada halaman admin	dapat menambah dan mengedit akun admin	95%	valid
9.	Login ke halaman alumni	Pengisian username dan password	masuk ke halaman admin/dashboard	95%	valid
10.	Logout dari halaman alumni	klik tombol logout pada navbar sebelah kanan atas	keluar dari halaman admin dan masuk ke halaman login pengisian username dan password	95%	valid
11.	masuk ke halaman profil Alumni	melakukan update data diri	update data diri alumni	95%	valid
12.	menisi surfey Tracer Study	klik tombol ikuti survey	dapat mengisi data survey	95%	valid

CONCLUSION

The research conducted at Dian Nusantara University reveals that the tracer study information system still relies on Google Forms. Based on these findings, the author developed a web-based tracer study information system. This system allows alumni to update their current personal information and complete the tracer study survey more flexibly. The application also facilitates the tracer study team in tracking and managing alumni data, enabling efficient survey completion monitoring and simplifying the analysis of survey responses from alumni.

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